

Cybersecurity & Ethical Hacking

BUILD YOUR PERSONAL CYBERSECURITY LAB

Task-2

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Objective

The objective of this task is to create a safe and isolated cybersecurity lab environment using Kali Linux as the attacker machine and OWASP Juice Shop as the vulnerable target application. The lab environment is used for practicing scanning, interception, packet analysis, and web application testing.

Lab Environment Configuration

Component	Configuration
Host System	Apple MacBook (M1, 8GB RAM, macOS Tahoe)
Virtualization Software	UTM
Attacker Machine	Kali Linux ARM64
Target Application	OWASP Juice Shop
Network Type	Shared Network / NAT

Step 1: Updating Kali Linux

Description:

The Kali Linux system was updated to ensure all packages and repositories were up to date.

Command Used:

sudo apt update && sudo apt upgrade -y

**Step 2: Creating Lab Directories****Description:**

Separate folders were created to store scans, notes, packet captures, and screenshots.

Command Used:

mkdir -p ~/lab/{scans,notes,pcaps,screenshots}

Step 3: Installing Required Tools**Description:**

Required cybersecurity tools such as Docker, Nmap, Wireshark, and Burp Suite were installed.

Command Used:

sudo apt install -y docker.io nmap wireshark burpsuite

Step 4: Starting Docker Service**Description:**

Docker service was enabled and started to deploy the vulnerable application.

Command Used:

sudo systemctl enable --now docker

docker --version

```
(kali㉿kali)-[~]
$ sudo systemctl enable --now docker

(kali㉿kali)-[~]
$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

Step 5: Deploying OWASP Juice Shop

Description:

OWASP Juice Shop vulnerable web application was downloaded and deployed using Docker.

Command Used:

```
sudo docker pull bkimminich/juice-shop
```

```
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

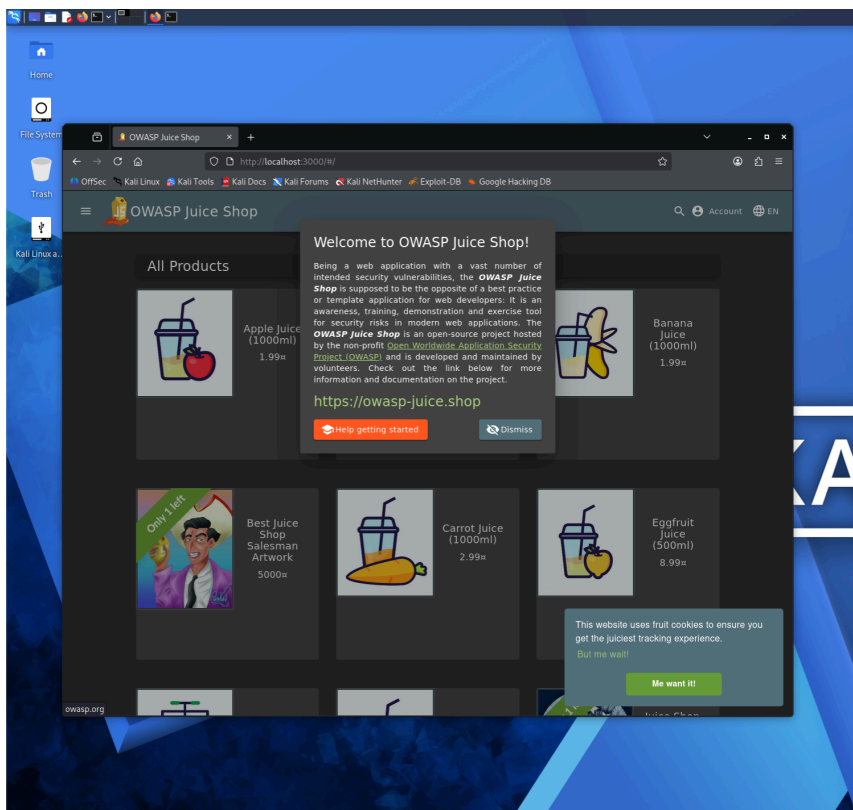
Step 6: Accessing the Vulnerable Application

Description:

The Juice Shop application was accessed through the web browser using localhost.

Command Used:

```
http://localhost:3000
```



Step 7: Verifying Network Configuration

Description:

The IP configuration of the Kali Linux VM was verified using the ip command.

Command Used:

ip a

Step 8: Nmap Scan Validation

Description:

Nmap was used to scan the local machine and verify open services and ports.

Command Used:

nmap localhost

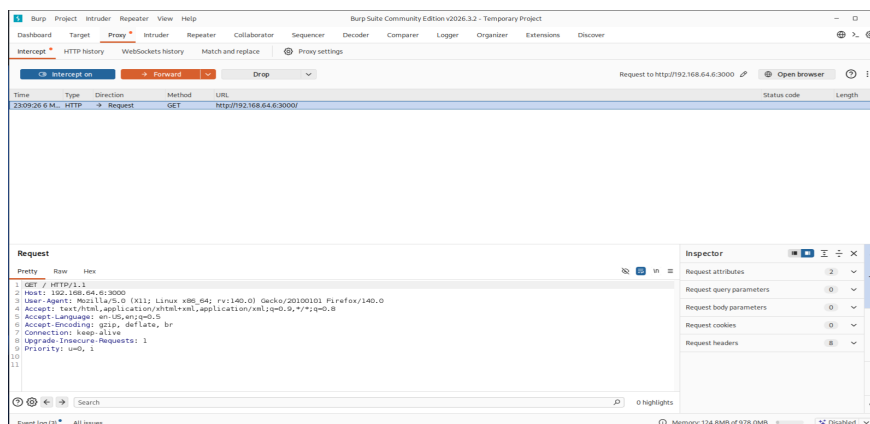
Step 9: Burp Suite Validation

Description:

Burp Suite was used to intercept HTTP requests between the browser and the vulnerable application.

Command Used:

Open Burp Suite → Temporary Project → Start Burp → Intercept ON



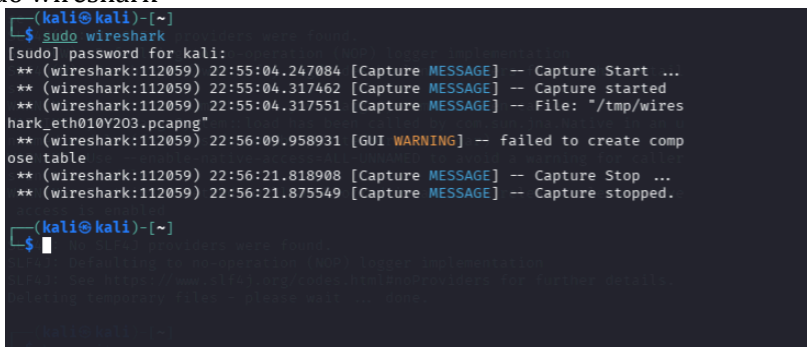
Step 10: Wireshark Validation

Description:

Wireshark was used to capture and analyze network packets generated by the Juice Shop application.

Command Used:

sudo wireshark



The cybersecurity lab environment was successfully created using Kali Linux and OWASP Juice Shop. The setup was validated using Docker, Nmap, Burp Suite, and Wireshark. This isolated environment can now be used for practicing ethical hacking and web application security testing.